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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
May 2025

CB2102-1- FUNDAMENTALS OF COMPUTER NETWORKS
(For CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which type of service does the network layer provide to the transport layer in connection-oriented service?
A) Reliable end-to-end delivery B) Datagram delivery
C) Broadcast delivery D) File transfer
2. Which routing method allows routers to learn dynamically about their neighbors?
A) Broadcast Routing B) Distance Vector Routing
C) Hierarchical Routing D) Multicast Routing
3. What is a feature of multicast routing?
A) Sends packets to a single destination B) Sends packets to multiple specific destinations
C) Requires tunneling D) Routes packets only in wired networks
4. Which congestion control approach involves predefining limits for traffic?
A) Load shedding B) Admission control
C) Traffic throttling D) Packet shaping
5. Which of the following is an example of integrated services in QoS?
A) RSVP (Resource Reservation Protocol) B) TCP/IP Protocol
C) Load Shedding D) DNS Services
6. What is the role of a virtual circuit in the network layer?
A) Provides a direct physical link B) Ensures guaranteed delivery of packets
C) Establishes a logical path for packet delivery D) Handles congestion control only
7. Which algorithm relies on knowledge of the entire network topology?
A) Flooding B) Link State Routing
C) Distance Vector Routing D) Shortest Path First
8. Which routing method sends packets to a group of specific hosts?
A) Anycast Routing B) Multicast Routing
C) Broadcast Routing D) Unicast Routing
9. Which type of routing is best suited for mobile hosts?
A) Broadcast Routing B) Routing in Ad hoc Networks
C) Hierarchical Routing D) Distance Vector Routing
10. What is a key characteristic of connectionless services?
A) Guaranteed delivery of packets B) Low latency in transmission
C) Packets may arrive out of order D) Requires session establishment

- Which congestion control technique drops packets to prevent overflow?**
11. A) Load shedding B) Traffic shaping
C) Admission control D) Packet scheduling
- Which routing method sends packets to a group of specific hosts?**
12. A) Anycast Routing B) Multicast Routing
C) Broadcast Routing D) Unicast Routing
- What is the key difference between connection-oriented and connectionless services?**
13. A) Connection-oriented requires session establishment, connectionless does not B) Connection-oriented is faster than connectionless
C) Connection-oriented is less reliable than connectionless D) Connectionless requires acknowledgments for packets
- Which of the following is NOT a design issue of the network layer?**
14. A) Packet forwarding B) End-to-end delivery
C) Routing algorithms D) Error detection
- Which of the following is a proactive routing algorithm?**
15. A) Distance Vector Routing B) Flooding
C) On-Demand Routing D) Hierarchical Routing
- What does the term "traffic-aware routing" refer to?**
16. A) Routing based on user behavior B) Routing that adjusts dynamically to traffic conditions
C) Routing with fixed paths for each flow D) Routing using static tables
- Which of the following is an example of traffic shaping?**
17. A) Token Bucket Algorithm B) TCP Sliding Window
C) Link State Routing D) Admission Control
- What is the purpose of admission control in congestion control?**
18. A) Limiting traffic entry into the network B) Prioritizing packets
C) Increasing bandwidth D) Dividing packets into smaller sizes
- What does the term "datagram subnet" refer to?**
19. A) A connection-oriented network B) A network that uses connectionless service
C) A network with fixed routes D) A network with guaranteed delivery
- In Link State Routing, each router maintains information about:**
20. A) The entire network topology B) Only its neighbors
C) The shortest paths to all nodes D) The congestion status of the network

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the process of congestion control using traffic throttling and load shedding.	8	L*2	1	1
	b) Compare virtual circuit and datagram subnets and discuss their applications.	8	L3	2	2
2.	a) Describe the shortest path routing with an example. How is Dijkstra's algorithm used in this context?	8	L2	1	1
	b) Differentiate between connectionless and connection-oriented services in the network layer with examples.	8	L3	2	2
3.	a) Define the concept of store-and-forward packet switching. Explain how it is implemented in a network.	8	L2	1	1
	b) Analyze the working of distance vector routing with a step-by-step example.	8	L3	2	2

Unit – II

- | | | | | | |
|-------|--|---|----|---|---|
| 4. a) | Define fragmentation in the network layer. Explain the process and how fragmented packets are reassembled. | 8 | L2 | 3 | 1 |
| b) | Discuss the structure of IPv4 and IPv6 addresses and list their differences. | 8 | L2 | 4 | 1 |
| 5. a) | Describe the role of OSPF and BGP in internetwork routing. | 8 | L2 | 3 | 1 |
| b) | Outline the services provided by the transport layer to the upper layers. | 8 | L2 | 4 | 2 |
| 6. a) | Explain the working of the User Datagram Protocol (UDP) with its key features and limitations. | 8 | L2 | 3 | 1 |
| b) | Write a short note on tunneling and its applications in network communication. | 8 | L2 | 3 | 2 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|---|
| 7. a) | Explain the TCP sliding window protocol with an example. | 8 | L2 | 5 | 1 |
| b) | Discuss the architecture of DNS and explain the types of records stored in DNS. | 8 | L2 | 5 | 1 |
| 8. a) | Explain the working of SMTP in electronic mail transfer and compare it with HTTP. | 8 | L2 | 5 | 1 |
| b) | Describe the TCP service model and explain how TCP manages reliable data transfer. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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Fourth Semester B.Tech (CBCS) Degree Examinations
May 2025

CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT
(For CS, CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Explain what is meant by **PRODUCT** with reference to one of the eight principles as per the **ACM/IEEE Code of Ethics**.
A) The product should be easy to use B) Software engineers shall ensure that their products and related modifications meet the highest professional standards possible
C) Software engineers shall ensure that their products and related modifications satisfy the client D) It means that the product designed/created should be easily available
2. Which of the following property does not correspond to a good **Software Requirements Specification (SRS)** ?
A) Verifiable B) Ambiguous
C) Complete D) Traceable
3. Requirements should specify _____ but not 'how' aspects of a software.
A) 'what' B) 'how'
C) 'why' D) 'which'
4. **Software Engineering** is an engineering discipline that is concerned with?
A) how computer systems work B) theories and methods that underlie computers and software systems
C) all aspects of software production D) all aspects of computer-based systems development, including hardware, software and process engineering
5. The major shortcoming of waterfall model is?
A) the difficulty in accommodating changes after requirement analysis B) the difficult in accommodating changes after feasibility analysis
C) the system testing D) the maintenance of system
6. Project risk factor is considered in which **SDLC** model?
A) Incremental Model B) Waterfall model
C) Prototyping model D) Spiral model
7. If requirements are easily understandable and defined then which model is best suited?
A) Spiral model B) Waterfall model
C) Incremental Model D) Prototyping model

8. Identify which phase is not available in SDLC?
 A) Coding B) Testing
 C) Maintenance D) Abstraction
9. Which model in system modeling depicts the dynamic behavior of the system?
 A) Context Model B) Behavioral Model
 C) Data Model D) Object Model
10. _____ and _____ diagrams of UML represent Interaction modeling.
 A) Use Case, Sequence B) Class, Object
 C) Activity, State Chart D) Class, State
11. Which of the following is a complementary approach to function-oriented approach?
 A) Object oriented analysis B) Object oriented design
 C) Structured approach D) Both Object oriented analysis and design
12. Sequence of messages is emphasized by which UML diagram?
 A) state chart B) sequence
 C) activity D) collaboration
13. Agile Software Development is based on?
 A) Incremental Development B) Iterative Development
 C) Linear Development D) Both Incremental and Iterative Development
14. According to Agile manifesto what carries more value?
 A) Individuals and interactions over processes and tools B) Individuals and interactions over people and technique
 C) Individuals and interactions over projects and tools D) Individuals and interactions over products and tools
15. Verifying that whether individual software components are functioning correctly and identifying the defects in them is objective of which level of testing?
 A) Integration Testing B) Acceptance Testing
 C) Unit Testing D) System Testing
16. _____ and _____ processes are concerned with checking that software being developed meets its specification and delivers the functionality expected by the people paying for the software.
 A) Verification and Validation B) Analysis and Design
 C) Code and Design D) Analysis and Code
17. If an organization is unsure of its cost estimate, it may increase its price by a contingency over and above its normal profit. This factor is called _____.
 A) Financial Health B) Cost estimate uncertainty
 C) Market opportunity D) Requirements volatility
18. _____ is created that records the work to be done, who will do it, the development schedule, and the work products.
 A) Decision B) Solution
 C) Project plan D) Strategy
19. _____ is not a risk management activity.
 A) Risk assessment B) Risk generation
 C) Risk control D) Risk monitoring
20. Quality Management is essential for project _____.
 A) failure B) degradation
 C) delay D) success

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Why is software engineering important? Also, briefly explain with a neat diagram the various activities of requirements engineering process(RE).	08	L*1	1	1

b)	Explain the incremental software process model with a neat diagram and discuss its merits and demerits.	08	L2	2	1
2. a)	Define requirement. Also, discuss various types of non-functional requirements with suitable example for any software system of your choice.	08	L1	1	1
b)	Explain with a neat diagram spiral software process model.	08	L2	2	1
3. a)	List the ACM/IEEE code of ethics that software engineers should follow to maintain professional ethical standards. Also, with example summarize an essential attributes of a good software.	08	L1	1	8
b)	Explain the waterfall software process model with a neat diagram and discuss its merits and demerits.	08	L2	2	1
Unit – II					
4. a)	Recall the principles of agile manifesto.	04	L1	4	1
b)	Explain test-driven development (TDD) with a suitable diagram.	04	L2	4	1
c)	Model the use-case diagram of a medical receptionist role in a mental health care patient monitoring system (MHC-PMS). Also, give the tabular description of transfer-data use case.	08	L3	3	2
5. a)	What are the advantages of pair programming?	04	L1	4	1
b)	Explain briefly different stages in an object-oriented design process (OOD).	06	L2	3	1
c)	Model the sequence diagram for MHC – PMS transfer data use-case.	06	L3	3	2
6. a)	Explain various forms of release testing with suitable example for each.	08	L2	4	1
b)	Model the state transition diagram for microwave oven operation. Also, provide tabular representation of various states of microwave oven.	08	L3	3	2
Unit – III					
7. a)	Show various factors affecting software pricing.	04	L1	5	1
b)	Interpret with suitable diagram about software review process.	04	L2	5	1
c)	Explain risk management process with suitable diagram.	08	L2	5	2
8. a)	Recall various advantages of a cohesive group.	04	L1	5	1
b)	Illustrate the need of any four static software product metrics.	04	L2	5	1
c)	Explain project scheduling process with suitable diagram. ✓	08	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

[illegible]

CS3005-1- MICROPROCESSOR AND EMBEDDED SYSTEMS
(For CS, CB)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

11. Set time interrupt from DOS interrupt will store the hour information in which register
 A) CH B) CL
 C) DH D) DL
12. In a program, a Macro is being called 'n' times. Then how many times is the machine code generated for the same?
 A) 1 time B) 'n' times
 C) 'n-1' times D) 'n+1' times
13. Which register in ARM holds the flags indicating the status of the processor?
 A) General-purpose register B) Program Counter (PC)
 C) Stack Pointer (SP) D) Program Status Register (PSR)
14. How does the 7-segment display represent numbers?
 A) Using rows and columns of LEDs B) By turning on specific segments of LEDs
 C) Through alphanumeric LCD cells D) By activating relays
15. Which built-in Arduino function is used for serial communication?
 A) digitalWrite() B) Serial.begin()
 C) analogRead() D) pinMode()
16. What is the primary design philosophy of ARM embedded systems?
 A) Complex Instruction Set Computing (CISC) B) Reduced Instruction Set Computing (RISC)
 C) Very Long Instruction Word (VLIW) D) Superscalar Processing
17. Which pin is used by the 8086 to indicate that an external device is ready to exchange data?
 A) READY B) INTR
 C) ALE D) HLDA
18. Which bit in the MOV instruction encoding specifies whether the data transfer is of a byte or a word?
 A) D (direction) bit B) MOD field bit
 C) W (word) bit D) REG field bit
19. During an interrupt response, which register is the first to be pushed onto the stack by the 8086?
 A) CS (Code Segment) B) IP (Instruction Pointer)
 C) FLAGS D) DS (Data Segment)
20. During a read cycle, when are the address lines present on the multiplexed bus?
 A) After the data is transferred B) During the WR signal activation
 C) When the ALE signal is active D) When the HLDA signal is active

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Identify the addressing modes of the instructions given below and justify the answer with clear explanation. i) MOV WORD PTR[SI], 50H ii) MOV DS:[1000H], 10H iii) MOV AX, Num[BX+DI]	6	L*3	1	2
	b) Write an Assembly level Program to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm.	6	L3	2	2
	c) Explain the use of instruction queue in 8086 architecture.	4	L2	1	1
2.	a) Describe the roles of various status flags and Control flags in Intel 8086 Microprocessor along with the flag register format. If the two numbers 1100H and 60C9H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer.	6	L3	1	1

- b) How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example. 6 L1 2 1
- c) Write an Assembly level Program to display your name in front of the message "what is your name". 4 L3 2 2
3. a) Explain the working of following 8086 instructions along with example for each:
i) AAM ii) DAA 6 L2 2 1
- b) Explain the following assembler directives with example:
i. ASSUME
ii. EQU
iii. DD 6 L2 1 1
- c) Assume that SI is 3000H and the content of 2FFFFH = 04H; content of 3000H = 4EH; Content of 3001H = 9FH. Which value is stored in AX register after execution of the statement MOV AX, [SI]. Justify your answer. 4 L3 2 2

Unit – II

4. a) Briefly explain the Embedded system software components. 8 L2 4 1
- b) Write an Embedded C program to control the speed of rotation of a stepper motor using potentiometer. 4 L3 4 2
- c) Explain the operation of the following DOS INT 21H functions:
i) Function 3AH
ii) Function 2CH 4 L2 3 1
5. a) With code snippet, explain Serial.begin(), Serial.available(), Serial.println() functions. 6 L2 4 1
- b) Differentiate between Macro and a Procedure. Write a procedure to read two digit BCD number from keyboard and store in BL register. 6 L3 4 1
- c) Compare and contrast RISC and CISC Architecture. 4 L2 3 1
6. a) With a basic layout diagram, explain the ARM program status register by highlighting the functionalities of different fields. 8 L2 4 1
- b) Write an Embedded C program to perform up-down counter on a 7-segment display using switches. 4 L3 4 2
- c) Write an Assembly Language Program to compute the factorial of a positive integer 'n' using a recursive procedure. 4 L3 3 2

Unit – III

7. a) Categorize the following signals into minimum mode signals and maximum mode signals. Write their roles.
(i) S2' to S0' (ii) RQ' / GT1' and RQ' / GT0' (iii) ALE (iv) DEN (v) HLDA 8 L1 5 1
- b) Explain the coding template for the MOV instruction of 8086 Intel microprocessor and Generate machine code for the following instructions of Intel 8086 microprocessor
MOV AL, [1000] [BX] 8 L3 5 2
8. a) List the 8086 signals having common functions in both minimum and maximum mode and explain any four signals in detail. 10 L1 5 1
- b) What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling. 6 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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May 2025

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

(For CS, IS, AM, CC, AD, CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- How many sub arrays does the quick sort algorithm divide the entire array into?
 A) One B) Two
 C) Three D) Four
- Binary Search works only on which type of arrays?
 A) Unsorted arrays B) Sorted arrays
 C) Arrays with distinct elements D) Arrays of integers
- Linear search algorithm, also known as ____
 A) Binary search B) Simple search
 C) Sequential search D) Difficult search
- In asymptotic analysis, what does it mean if a function is said to be $O(n)$?
 A) The time complexity increases exponentially as n increases B) The algorithm's runtime grows linearly with the input size
 C) The runtime of the algorithm is independent of the input size D) The algorithm's runtime grows logarithmically as n increases
- What will be the output of the following pseudocode if the driver is fun(5)? function fun(n) if $n == 1$ return 1 else return $n + \text{fun}(n-1)$
 A) 120 B) 24
 C) 6 D) 15
- Pseudocode is written using
 A) Combination of Native & programming language B) High level Programming language only
 C) Natural language only D) Low level programming language
- Which of the following is NOT a characteristic of a good algorithm?
 A) Clarity and simplicity B) Efficiency in terms of time and space
 C) Correctness D) The use of complex data structures
- What is the primary characteristic of the Divide and Conquer approach?
 A) Solving problems by iteration B) Solving problems by dividing them into overlapping sub problems
 C) Dividing a problem into smaller independent subproblems, solving them recursively, and combining their results D) Dividing a problem into smaller problems and linearly solving it.
- Which technique is used to solve the 0/1 Knapsack problem efficiently?
 A) Greedy algorithm B) Divide and conquer
 C) Dynamic programming D) Depth-first search

10. What is the primary purpose of the Floyd-Warshall algorithm?
 A) Finding the shortest path between two specific nodes
 B) Finding the shortest paths between all pairs of nodes in a graph
 C) Finding the minimum spanning tree of a graph
 D) Finding the longest path in a weighted graph
11. In Horspool's algorithm, which table is used to store the shift values for the pattern matching process?
 A) Shift Table
 B) Hash Table
 C) Match Table
 D) Character Table
12. What makes Counting Sort a distribution-based algorithm?
 A) It uses element frequencies to determine sorted positions.
 B) It divides the array into smaller subarrays.
 C) It recursively splits and merges data.
 D) It compares each pair of elements to sort the array.
13. You are given an array: [3, 9, 2, 1, 4, 8]. After building a max-heap from this array, what will be the root element?
 A) 1
 B) 9
 C) 2
 D) 8
14. What operation is performed on an AVL tree when the balance factor of a node becomes greater than 1 or less than -1?
 A) Rotation
 B) Deletion
 C) Insertion
 D) Search
15. Binary Search is a classic example of which variation of Decrease and Conquer?
 A) Decrease by a constant
 B) Decrease by a constant factor
 C) Decrease by one
 D) Divide and conquer
16. In a max-heap, which of the following is true?
 A) The parent node is smaller than its child nodes
 B) The parent node is greater than or equal to its child nodes
 C) The tree is a binary search tree
 D) The elements are stored in ascending order
17. In the Job Assignment Problem, what does the objective function typically aim to minimize?
 A) The time to complete all tasks
 B) The number of tasks assigned
 C) The total cost of assigning tasks to workers
 D) The total time worked by all workers
18. In the subset-sum problem, when do we backtrack?
 A) When the sum of the selected elements exceeds the target sum
 B) When the sum of the selected elements is equal to the target sum
 C) When all elements have been considered
 D) When a subset has been found
19. Dijkstra's Algorithm is the prime example for _____
 A) Greedy algorithm
 B) Branch and bound
 C) Back tracking
 D) Dynamic programming
20. Which of the following algorithms is the best approach for solving Huffman codes?
 A) exhaustive search
 B) greedy algorithm
 C) brute force algorithm
 D) divide and conquer algorithm

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the Performance Analysis of an algorithm.	8	L*2	1	1
b)	Write the pseudocode to sort the elements in an array using selection sort and analyze its efficiency. Apply selection sort to sort the following elements in an array. 5, 4, 2, 1, 6, 3.	8	L3	2	2
2. a)	Write the general plan for analyzing the efficiency of non recursive algorithm. Write the pseudocode to find the maximal element in an array and analyze its efficiency.	8	L3	1	2

- b) Write Divide And Conquer recursive Merge sort algorithm and derive the time complexity of this algorithm.
3. a) Write the algorithm for brute force string matching. Show the comparisons the Brute force string matching makes for the pattern $P=acbcac$ in the text $T=acbcabccababcaacbcac$.
- b) Illustrate Quick sort algorithm. Sort the following list using Merge sort. 38, 27, 43, 3, 9, 82, 10.

Unit – II

4. a) Write an insertion sorting algorithm using decrease and conquer technique. Apply the algorithm on the following numbers and trace the steps - 23, 1, 10, 5, 2
- b) Construct AVL tree for the following input by showing each step of the insertion process and type of rotation used for transformation. 63, 9, 19, 27, 18, 108, 99, 81
5. a) Identify the pattern "BARBER" in the given text "JIM_SAW_ME_IN_A_BARBERSHOP" using Horspool's string-matching algorithm. Show the construction of shift table for the same.
- b) Write an algorithm for sorting list of integers by comparison method that uses input enhancement design technique. Apply the algorithm on the following numbers and trace the steps 25, 45, 10, 20, 50, 15.
6. a) Define topological sorting. Apply DFS based method to topologically sort the vertices of the following graph.

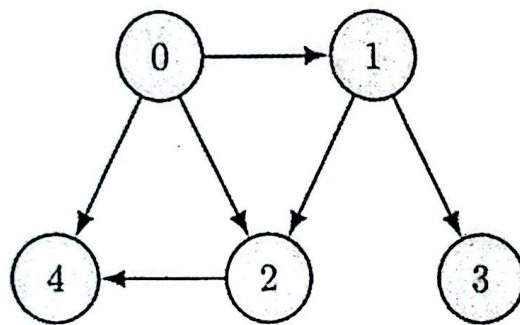


Fig. 6a

- b) Write and Apply Floyd's all pairs shortest path algorithm for the graph shown in the following figure.

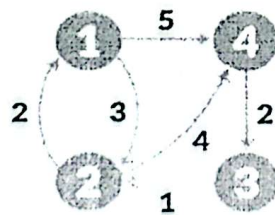


Fig. 6b

Unit – III

7. a) Solve the given instance of sum of subset problem $S=\{5, 10, 12, 13, 15, 18\}$ and $d=30$. Construct a state space tree.
- b) Define minimum cost spanning tree. Write Prim's algorithm to find minimum cost spanning tree.

8 L3 2 2

8 L3 1 2

8 L3 2 2

8 L3 3 2

8 L3 4 2

8 L3 3 2

8 L3 4 2

8 L3 3 2

8 L3 4 2

8 L3 5 2

8 L2 5 1

8. a) Explain the following: (i) Class P (ii) Class NP (iii) NP Complete Problem (iv) NP Hard Problem.
- b) Solve the job assignment problem using Branch and Bound technique for the following matrix.

8 L2 5 1

	Job 1	Job 2	Job 3	Job 4
A	9	2	7	8
B	6	4	3	7
C	5	8	1	8
D	7	6	9	4

Fig. 8b

8 L3 5 2

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